Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · MIXED ANOVA

between-groups × repeated conditions

Table 1. Descriptives by group × condition

Group	Condition	N	M	SD
High	1	20	7.41	0.46
Low	1	20	7.39	0.46
High	2	20	7.17	0.47
Low	2	20	6.78	0.51
High	3	20	6.71	0.49
Low	3	20	5.68	0.51

Table 2. Mixed ANOVA

Source	SS	df	MS	F	p	partial η^2 [95% CI]
Tolerance_group (between)	6.91	1	6.91	10.37	.003	0.21 [0.05, 1.00]
Condition (within)	29.83	1.60	18.60	759.23	<.001	0.95 [0.94, 1.00]
Tolerance_group × Condition	5.18	1.60	3.23	131.84	<.001	0.78 [0.70, 1.00]

Table 3. Sphericity (Mauchly's test)

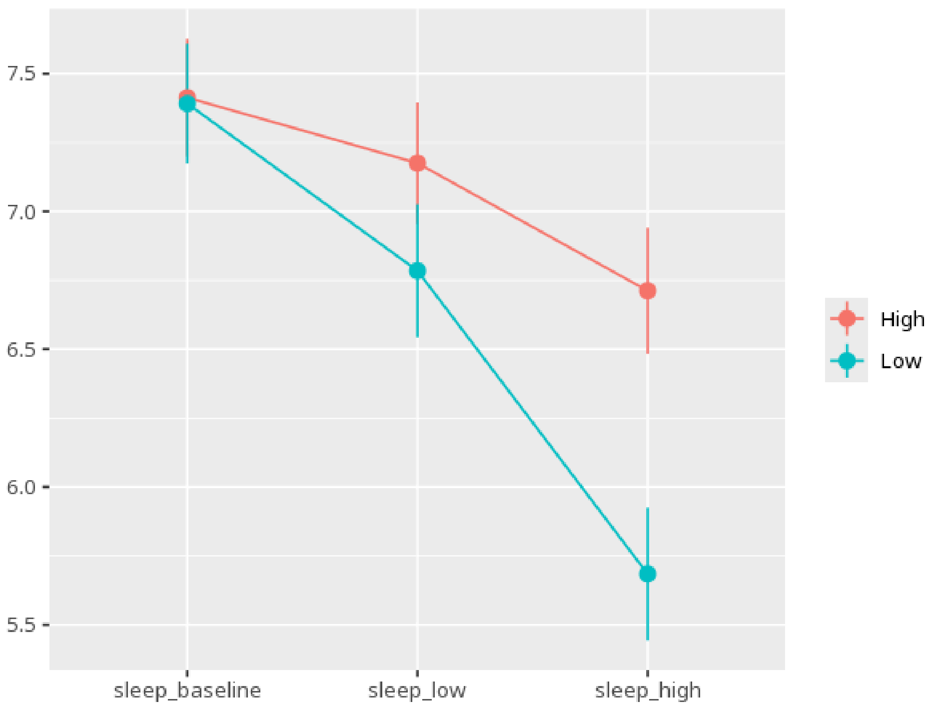
Effect	W	p	GG ε	HF ε
condition	0.75	.005	0.80	0.83
grp:condition	0.75	.005	0.80	0.83

assumption check: Levene's (equal variances between groups). between and within effects are tested against different error terms; when sphericity is violated the within and interaction F-tests use the Greenhouse–Geisser / Huynh–Feldt correction. Mauchly's test & the GG/HF corrections apply only when the repeated factor has 3+ levels (with 2 levels sphericity is automatically met and this table is omitted). (Levene F=0.23, p=.632) - equal variances between groups look reasonable - sphericity looks violated; check that a Greenhouse–Geisser or Huynh–Feldt correction is applied above

Table 4. Post-hoc comparisons (condition pairs)

Pair	M_{diff}	SE	p_{adj}	95% CI
sleep_baseline - sleep_low	0.42	0.03	<.001	[0.36, 0.49]
sleep_baseline - sleep_high	1.20	0.03	<.001	[1.13, 1.27]
sleep_low - sleep_high	0.78	0.04	<.001	[0.69, 0.88]

Figure. Means across conditions by group



Understanding the numbers

Condition. Each row names one combination of a between-groups level and a repeated-measures condition. Here, the descriptives table above reports 6 such group × condition combinations.

N. How many subjects contributed a value for that group × condition combination. Here, each of the 6 rows above lists its own N in the descriptives table.

M. The average score for that group × condition combination. Here, each row's mean is listed separately in the descriptives table above.

SD. How spread out scores are within that group × condition combination. Here, each row's SD is listed separately in the descriptives table above.

Source. Names which effect the row tests - the between-groups factor, the within-subjects condition factor, or their interaction. Here, the headline effect discussed above is Tolerance_group × Condition.

SS. The variability in the outcome attributable to that effect. Here, Tolerance_group × Condition has SS = 5.18.

df. The degrees of freedom for that effect, Greenhouse-Geisser/Huynh-Feldt-corrected for the within and interaction terms when sphericity is violated. Here, Tolerance_group × Condition has df = 1.60.

MS. The effect's SS divided by its df. Here, Tolerance_group × Condition has MS = 3.23.

F. The ratio of that effect's mean square to its error term - between and within effects use different error terms. Here, F for Tolerance_group × Condition = 131.84.

p. The probability of an effect this large if that factor, condition, or interaction truly had no influence. Here, $p < .001$ for Tolerance_group × Condition - below the $\alpha = 0.05$ threshold.

partial η^2. The proportion of variance explained by that effect. Here, partial η^2 for Tolerance_group × Condition = .78 [.70, 1.00].

Effect (sphericity). Names which within-subjects or interaction term Mauchly's test is checking for sphericity. Here, sphericity is checked for condition.

Mauchly's W. How far that term departs from the sphericity assumption (closer to 1 = more spherical). Here, W = 0.75.

GG ε. The Greenhouse-Geisser correction factor applied when sphericity is violated. Here, GG ε = 0.80.

HF ε. The Huynh-Feldt correction factor - an alternative, usually less conservative, sphericity adjustment. Here, HF ε = 0.83.

M diff. The difference between two condition means in the post-hoc table. Here, each post-hoc row above reports its own mean difference between a pair of conditions - but read each group's own line on the profile plot when the interaction is significant.

SE. The standard error of that mean difference. Here, each contrast's SE is listed alongside its mean difference in the post-hoc table above.

p (adj). The pairwise p-value, adjusted for the number of condition comparisons made. Here, each pairwise comparison above carries its own multiple-comparison-adjusted p.

95% CI. The range likely to contain the true mean difference for that pair of conditions. Here, each contrast's 95% CI is shown in the post-hoc table above.

How to read this test

Read the Group × Condition interaction first - a significant interaction means the groups changed differently across conditions; only then read the main effects. Check sphericity for the within and interaction terms - if violated, read the corrected F/p. When the group × condition interaction is significant, the overall condition comparisons in the post-hoc table can mislead - read each group's line on the profile plot instead. Your significance threshold (α) is 0.05.

APA template: A mixed ANOVA yielded a Tolerance_group × Condition interaction, $F(1.60,60.96)=131.84$, $p < .001$, partial $\eta^2=.78$ [.70, 1.00].

Statistical basis: Mixed (split-plot) ANOVA (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Mauchly's test of sphericity (rendered above the correction) (Mauchly, J. W. (1940). "Significance test for sphericity of a normal n-variate distribution." The Annals of Mathematical Statistics, 11(2), 204-209.); Greenhouse-Geisser correction (sphericity violated) (Greenhouse, S. W., Geisser, S. (1959). "On methods in the analysis of profile data." Psychometrika, 24(2), 95-112.); Huynh-Feldt correction (sphericity violated) (Huynh, H., Feldt, L. S. (1976). "Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs." Journal of Educational Statistics, 1(1), 69-82.); Levene's test for equal variances between groups (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), Contributions to Probability and Statistics (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." Journal of the American Statistical Association, 69(346), 364-367.); Estimated marginal means for post-hoc comparisons (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.). Full references in the exported CITATIONS.txt.

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